Project	**Mansourah Massarah Gold Project**
Customer:	**MA'ADEN GOLD AND BASE METALS (MA'ADEN)**

Title:	**GENERAL ARRANGEMENT DRAWING 49PR POWER GENERATION OILY WATER BUFFER TANK 50m3**

Document Number:	**MD-216-49PR-EG-ME-GAD-V553-1038**
Revision:	**D01**
Revision Date:	**17.11.2023**
Status:	**AS-BUILT**

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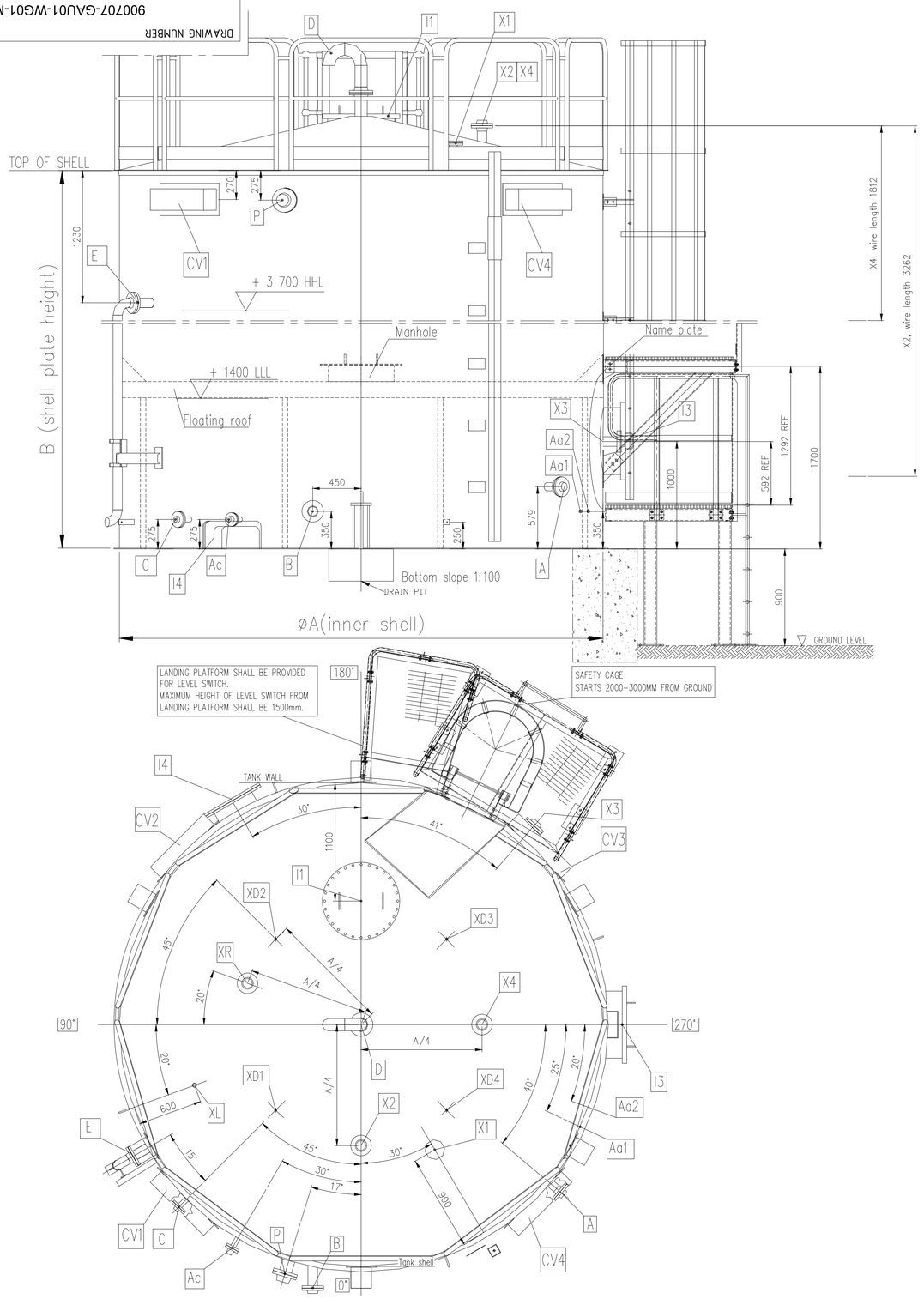
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REVISION HISTORY

Rev	Date	Revision Text	Prepared By	Reviewed By	Approved By	Employer
A01	01.04.2020	FIRST ISSUE	Wartsila	SJK / DBH	NBR	
A02	22.06.2020	REVISED AS PER EMPLOYER COMMENTS DATED 27.04.2020	Wartsila	SJK / DBH	NBR	
A03	09.10.2020	REVISED AS PER EMPLOYER COMMENTS DATED 14.07.2020	Wartsila	SJK / DBH	NBR	
C01	12.11.2020	ISSUED FOR CONSTRUCTION	Wartsila	SJK / DBH	NBR	
C02	23.08.2021	ISSUED FOR INFORMATION. REVISED DUE TO HFO TO ALCO CONVERSION	Wartsila	SJK	NBR	
C03	28.10.2021	ISSUED FOR CONSTRUCTION	Wartsila	SJK	NBR	
D01	17.11.2023	AS-BUILT	NAEW	SJK	NBR	

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REV D01



NOZZLE TABLE

MARK	DESCRIPTION	SIZE	FLANGE	ORIENTATION	PROJ. (min)	ELEVATION	REMARKS
A	FROM OILY WATER COLLECTING SUMPS	3" / DN80	x	310°	175	579	
B	TO OILY WATER TREATMENT	3" / DN80	x	450 FROM 0°	175	350	
C	DRAIN / EMPTYING	1½" / DN40	x	45°	150	275	
D	VENTING PIPE	4" / DN100	x	AT CENTER	150	ROOF	WIRE MESH
E	OVERFLOW	3" / DN80	x	60°	175	1230 FROM T.O.S.	
I1	ROOF MANHOLE / EMERGENCY VALVE	24" / DN600	x API 650	180°	250	ROOF	
I3	SHELL MANHOLE	24" / DN600	x API 650	270°	150	1000	
I4	CLEANOUT DOOR	203X406		150°		0	See Table 5.9a, 5.10a, 5.11a in API650
Aa1	TEMPERATURE LOCAL	G1/2"		295°		350	
Aa2	TEMPERATURE SENSOR (OPTIONAL / PLUGGED)	G1/2"		290°		350	
P	FOAM POURER (OPTIONAL / BLIND FLANGE)	4" / DN100	x	17°	175	275 FROM T.O.S.	
X2	HIGH LEVEL CONTROL (BLIND FLANGE)	3" (150 lbs)	x	0°	150	ROOF	
X3	LOW LEVEL CONTROL	3" (150 lbs)	x	221°	175 MAX	1000	
X4	HIGH HIGH / HIGH LEVEL CONTROL	3" (150 lbs)	x	270°	150	ROOF	
Ac	CONN.FOR PRESSURE TRANSM.	1 1/2"(150 lbs)	x	30°	150	275	
X1	CONNECTION FOR LOCAL LEVEL INDICATOR	2½" / DN65	x	330°	150	ROOF	
XR	ALT. CONN. FOR LEVEL MEASUREMENT	3" (150 lbs)	x	110°	150	ROOF	
XL	ALT. CONN. FOR HIGH HIGH-LEVEL CONTROL(PLUGGED)	G1"		70°		ROOF	
XD1-XD4	HEAT DETECTORS (OPTIONAL / PLUGGED)	1/2" 14NPT		45°, 135°, 225°, 315°		ROOF	TOTAL 4 PCS
CV1 - CV4	CIRCULATION VENT	700 x 200		45°, 135°, 225°, 315°		270 FROM T.O.S.	
	MANHOLE						LOCATION IN FLOATING ROOF
x ADDITIONAL INFO FROM API 650 DATA SHEETS							

ACCESSORIES

DESCRIPTION	SIZE	QTY.	REMARKS
ROOF HANDRAIL	H= 1200 mm	1 SET	
NAME PLATE	250 x 180	1 SET	ABOVE MANHOLE
LADDERS		1 SET	
EARTHING BRACKET	12 x 50 x 150	4 SETS	
PLATFORM		1 SET	
ANCHORING FOOTS			ACC TO PROJECT SPEC. TANKLIST
EMERGENCY VALVES ACC TO API 2000		1 SET	
DRAIN PIT	24" / DN600	1 SET	ACC. TO API, CENTER UOS.
x ADDITIONAL INFO FROM API 650 DATA SHEETS			

DESIGN CONDITIONS & MATERIALS
SEE TANK DATA SHEETS

FLOATING ROOF

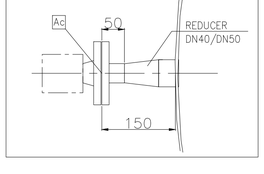
- ANTI ROTATION CABLES
- HONEYCOMB TYPE FLOATING ROOF
- ANTISTATIC CABLE
- DOUBLE SEAL

MEASURES			VOLUMES		PRELIMINARY DESIGN VALUES		
Nominal size	ø A (mm)	B (mm)	Net (m3) *3)	Gross (m3) *4)	* Weight,Dry (kg)	Strength calc.	Data Sheet
50m3 / 315bbl	4500	5000	36.6	59	8277	DBAF502231	DBAF502283

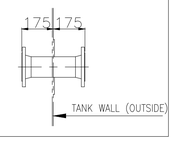
** Including corrosion allowance 1 mm (2mm for bottom plate)
*3) Between the floating roof elevations +1400 ... +3700
*4) From tank bottom to elevation +3700

Plates (mm)	50m3
Roof plate	6 **
Course 3	6 **
Course 1-2	8 **
Bottom plate	8 **

CONNECTION FOR PRESSURE TRANSMITTER



TYPICAL SIDE VIEW OF NOZZLE - P



NOTE:

- API DATA SHEET TO BE FILLED IN EVERY CASE.
- FOR THE CIVIL DESIGN INPUT WEIGHT FROM PRELIMINARY STRENGTH CALCULATIONS.
- ALL DIMENSIONS ARE IN MILLIMETER.
- COUNTER FLANGES INCLUDED (WELDNECK).
- *FINAL WEIGHT OF THE TANK AND SPECIFIED STRENGTH CALCULATIONS BY TANK MANUFACTURER.
- SURFACE TREATMENT SEE DOCUMENT WDAAA203188 LATEST REVISION
- TANK TESTING ACC. TO API 650 (LATEST VERSION)
- DOCUMENT MANAGEMENT SYSTEM AND DOC ID HAS CHANGED
- DBAF501459 'e' HAS BEEN REPLACED BY C12164-0189

REV	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
		D01	17-11-2023	AS-BUILT	NAEW	MNP	--	QRS	
QRS		A01	01-04-2020	ISSUED FOR INFORMATION	--	MNP	--	QRS	
DOCUMENT NUMBER									

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DRAWING NUMBER					REVISION
C12164-0189					B
DATE	03-Jun-2021	DATE	10-Jun-2021	DATE	10-Jun-2021
DRAWN	UKAX01	CHECKED	SINX04	APPROVED	GBH002

LARSEN & TOUBRO DRAWING NUMBER	REVISION
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DRAWN	--	--
CHECKED	MNP	27/03/2020
ENGINEER	--	--
APPROVED	QRS	01/04/2020
EMPLOYER		
SCALE	1:XX	A1

MANSOURAH - MASSARAH GOLD PROJECT
Dimensional Drawing
Power Generation
Oily Water Buffer Tank 50m3

DRAWING NUMBER	REV
MD-216-49PR-EG-ME-GAD-V553-1038	D01

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